

EPS 522/ENVS 423L Problem Set #4

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Calculations, methods, and assumptions should be spelled out so that your work can be reproduced. All work should be shown to receive full credit.

Upload a word document or PDF of your solutions/answers by midnight on Thursday, September 25 to Canvas.

1 Introduction

The purpose of this problem set is to reinforce your understanding of snow hydrology and evapotranspiration (ET).

1.1 Snow Water Equivalent (SWE)

Snow surveyors using a snow tube (with an interior diameter of 3.756 cm) recorded the following data from a snow course on two different dates. Calculate the SWE for each date, assuming the scale has been tared to the weight of the snow tube, such that the reported weight is only for the snow. (15 pts)

Date	Snow Depth (m)	Weight (g)	SWE (cm)
March 2	.8	330	
April 1	.9	415	

1.2 SWE Metrics

Using the included `taos_pwdernhorn_snotel.csv` data, plot SWE for water year 2025. Calculate the snow cover duration, accumulation rate, peak SWE amount, peak SWE date, ablation rate, and melt-out date for the station. Note that the accumulation and ablation rates are averages covering the start of snow cover to peak SWE and peak SWE to melt out date. (15 pts)

1.3 Pan Evaporation

Calculate the cross-sectional area and storage volume in square meters for a Class A evaporation pan. (5 pts)

Calculate the depth of water in the pan, if the pan contains 10 US gallons of water. (5 pts)

After 24 hours in an open field the evaporation pan now has 9.25 gallons of water. There was no precipitation during this period. Calculate the average evaporation rate from the pan in mm/hr. (5 pts)

Assume the pan was refilled to a volume of 10 gallons and again left in the open field for 24 hrs. During this time precipitation fell for 3 hrs with an average intensity of 2.5 mm/hr. Ending volume in the pan was 11.5 gallons. What was the amount of evaporation? (10 pts)

Calculate the flux of latent heat from the water in the pan to the atmosphere in the question where the ending volume in the pan was 9.25 gallons. Assume that the latent heat of vaporization is 2.45×10^6 J/kg and the density of water is 1000 kg/m³. (10 pts)

1.4 Actual ET

Go to the openET webpage, create an account if you haven't yet (sorry! but also I use a burner email account for things like this if you don't think you'll use it...). Click "explore data" and click "gridded data" when the option comes up. Pick a location and plot the cumulative ET (include a screen grab of the plot below). (10 pts)

What is the mean and approximate uncertainty of the ensemble cumulative ET value? (hint: spread from mean) (10 pts)

Pick one of the models used in the openET ensemble and describe the approach to quantifying ET, the input datasets, and sources of uncertainty for the approach. (15 pts)